

```
1 BEGIN{
2     sum =0
3     count = 0
4     print "Processing Employee Records.."
5 }
6 $2 > 6000 && $3 > 4{
7     sum+=$2
8     count++
9 }
10 END{
11     if(count>0){
12         print "Average Pay: ",sum/count
13         print "Sum: ",sum
14         print " Number of employees: ",count
15     }
16     else{
17         print "No employees match the criteria"
18 }
19 }
```

```
1 BEGIN{
2     print "Name sub-1 sub-2 sub-3 status"
3     print "-----"
4 }
5 {
6     status = "pass"
7     for(i =2; i <= NF; i++){
8         if($i < 45 ){
9             status = "fail"
10            break
11        }
12    }
13    print $0,status
14 }
15 END{
16     print "-----"
17 }
18
```

Open ▼



employees.dat

```
1 Joe 8000 5
2 Ram 6000 4
3 Tim 3000 3
```

```
[dhyanesh@localhost ~]$ touch average.awk
[dhyanesh@localhost ~]$ vi average.awk
[dhyanesh@localhost ~]$ touch employees.dat
[dhyanesh@localhost ~]$ vi employees.dat
[dhyanesh@localhost ~]$ awk -f average.awk employees.dat
Processing Employee Records..
Average Pay: 8000
Sum: 8000
Number of employees: 1
[dhyanesh@localhost ~]$ touch mark.awk
[dhyanesh@localhost ~]$ vi mark.awk
[dhyanesh@localhost ~]$ touch subject.dat
[dhyanesh@localhost ~]$ vi subject.dat
[dhyanesh@localhost ~]$ awk -f mark.awk subject.dat
Name sub-1 sub-2 sub-3 status
-----
1 Ben 30 45 50 fail
2 Tim 99 96 97 pass
3 Tom 96 94 94 pass
   pass
-----
[dhyanesh@localhost ~]$ vi subject.dat
[dhyanesh@localhost ~]$ ~
bash: /home/dhyanesh: Is a directory
[dhyanesh@localhost ~]$ █
```



```
#!/bin/bash
```

```
echo "Enter a Year"
```

```
read year
```

```
if [ $((year % 400)) -eq 0 ] || { [ $((year % 4)) -eq 0 ] && [ $((year % 100)) -ne 0 ] }  
then
```

```
    echo "$year is a leap year"
```

```
else
```

```
    echo "$year is not a leap year"
```

```
[dhyanesh@localhost shell_lab]$ chmod +x leapyear.sh
```

```
[dhyanesh@localhost shell_lab]$ ./leapyear.sh
```

```
Enter a Year
```

```
2024
```

```
2024 is a leap year
```

Open ▼



subject.dat

~/

Ben 30 45 50

Tim 99 96 97

Tom 96 95 99

```
dhyanesh@localhost:~$ touch exp2.cd
```

```
dhyanesh@localhost:~$ vi exp2.cd
```

```
dhyanesh@localhost:~$ bash exp2.cd
```

```
Enter number 1:
```

```
5
```

```
Enter number: 2
```

```
10
```

```
1. Add      2. Subtract      3. Multiply      4. Division
```

```
1
```

```
Result: 15
```